

ENGINEERING MATERIALS

ME1031 – Engineering Skill Development

Department of Mechanical Engineering, Sri Lanka
Institute of Information Technology



Picture courtesy: <https://www.orbitlanka.com/recycled-plastic-pellets/>

Presented by Dr. L U Subasinghe | PhD (SG), BSc Eng Hons (SL), CEng.MIE(SL), AMIMechE
Senior Lecturer

Department of Mechanical Engineering, University of Moratuwa
lihils@uom.lk

Learning outcomes

- At the end of this section, students should be able to
 - LO-6: Use of basic hand tools and machine tools in engineering workshop for machining processes and joining metals
 - **Materials used for such manufacturing processes**

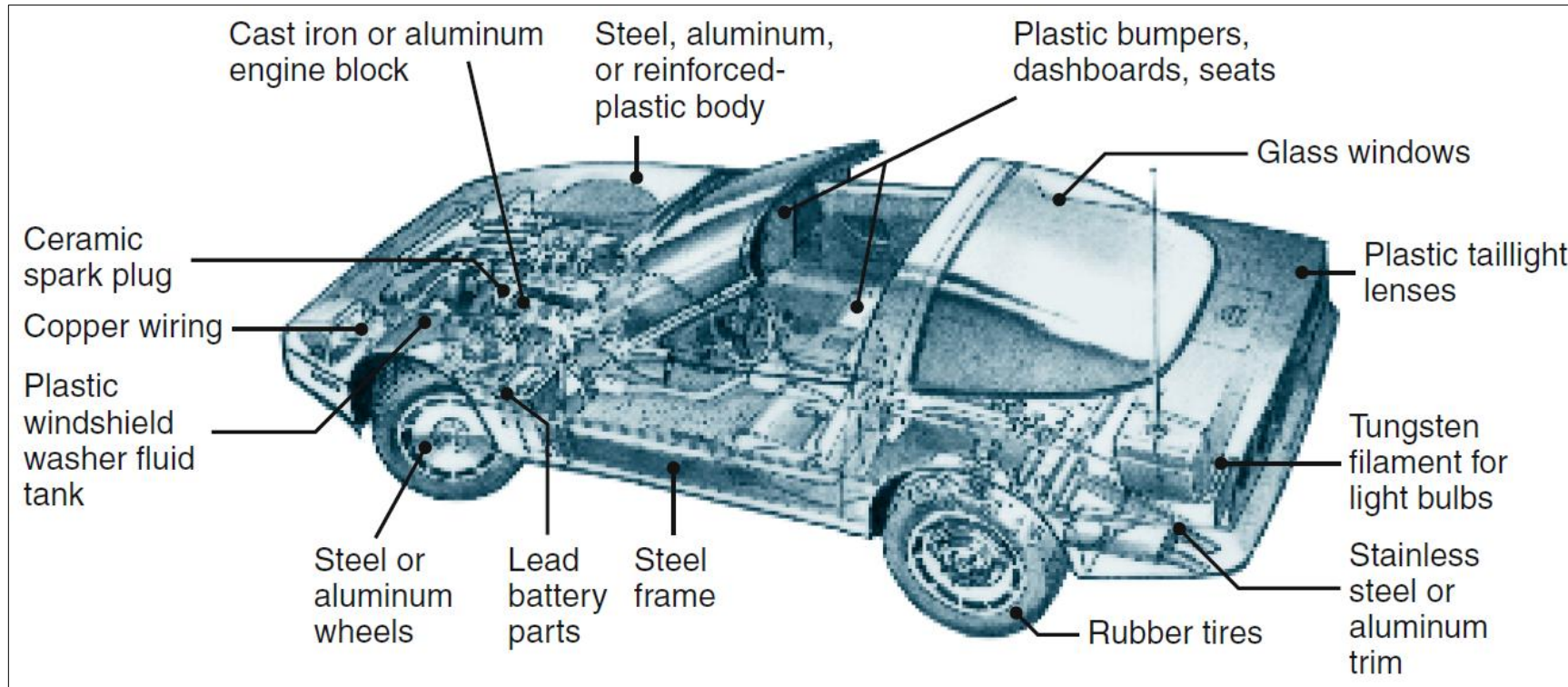
Outline

- What are engineering materials?
- Selection of materials for different applications
- Relationship between materials, properties and manufacturing processes
- Material failure
- Cost of materials

WHAT ARE ENGINEERING MATERIALS?

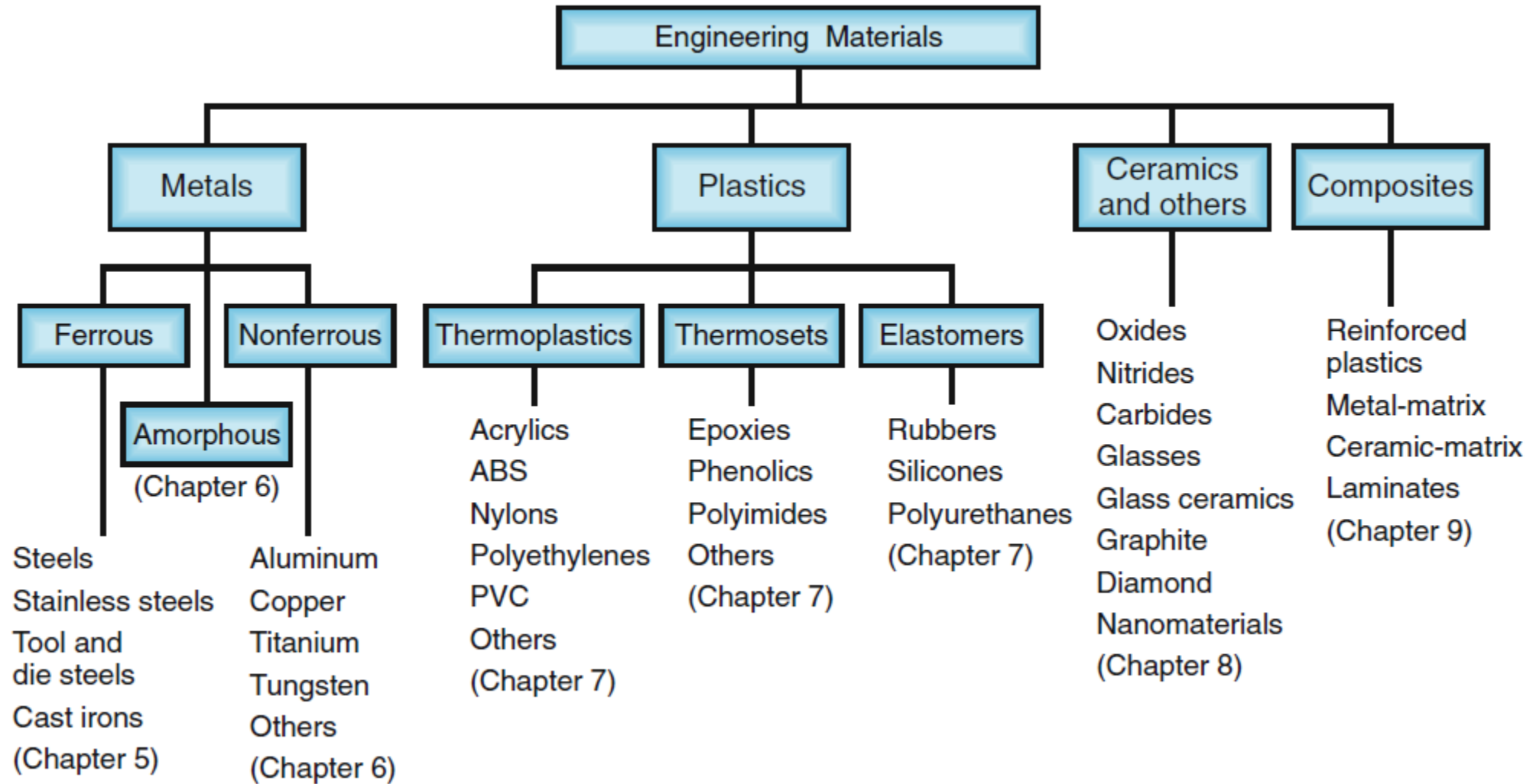
WHAT ARE ENGINEERING MATERIALS?

- Materials that are used in engineering applications
 - These applications can vary between a small component like a light bulb to a large skyscraper



Some of the metallic and nonmetallic materials used in a typical automobile

ENGINEERING MATERIALS CLASSIFICATION



STANDARD FORMS AND SIZES

- Metals

- Billets, Bars, Foils (thickness < 0.2 mm), Sheets (thickness < 6 mm), Plates, Tubes, Powders

- Plastics

- Powders, Pellets

- Ceramics

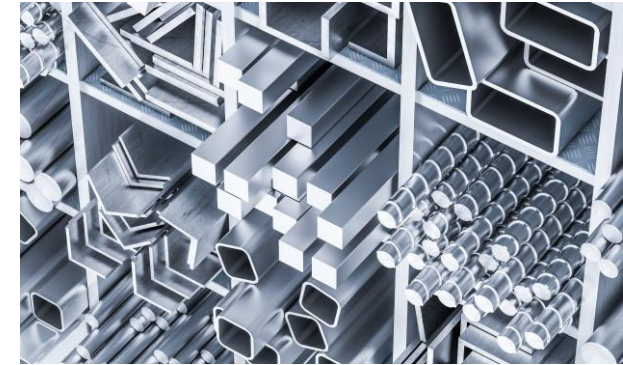
- Powders (liquids are also required in the slurry preparation process)

- Composites

- Powders (liquids are also required in the slurry preparation process)



Metal billet



Metal bars



Plastic powders



Ceramic powders

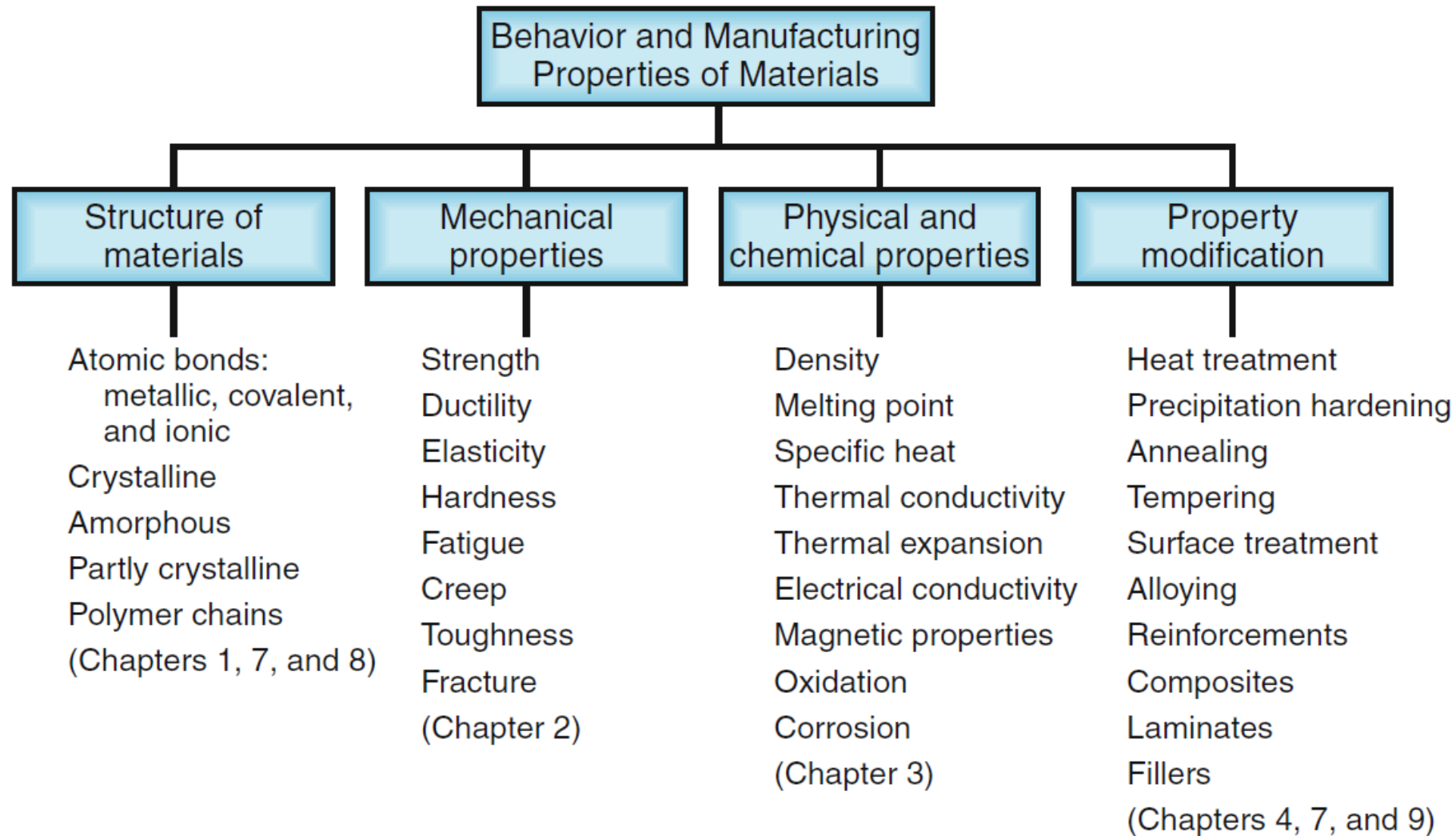
Kalpakjian, S., & Schmid, S. R. (2006). Manufacturing Engineering and Technology (6th ed.). Pearson Education South Asia.
<https://sciencing.com/difference-between-sheet-plate-steel-7798826.html>
<http://www.kortsa.com/pages/lead-billets.html>
https://www.freepik.com/premium-photo/metal-bar-3d_4482608.htm
<http://corofin-cn.com/plastic-products/>
<https://www.kennametal.com/cn/zh/products/sintec-ceramics/ceramic-powders.html>
<https://www.youtube.com/watch?v=sq3KMEG7z0g>

SELECTION OF MATERIALS FOR DIFFERENT APPLICATIONS

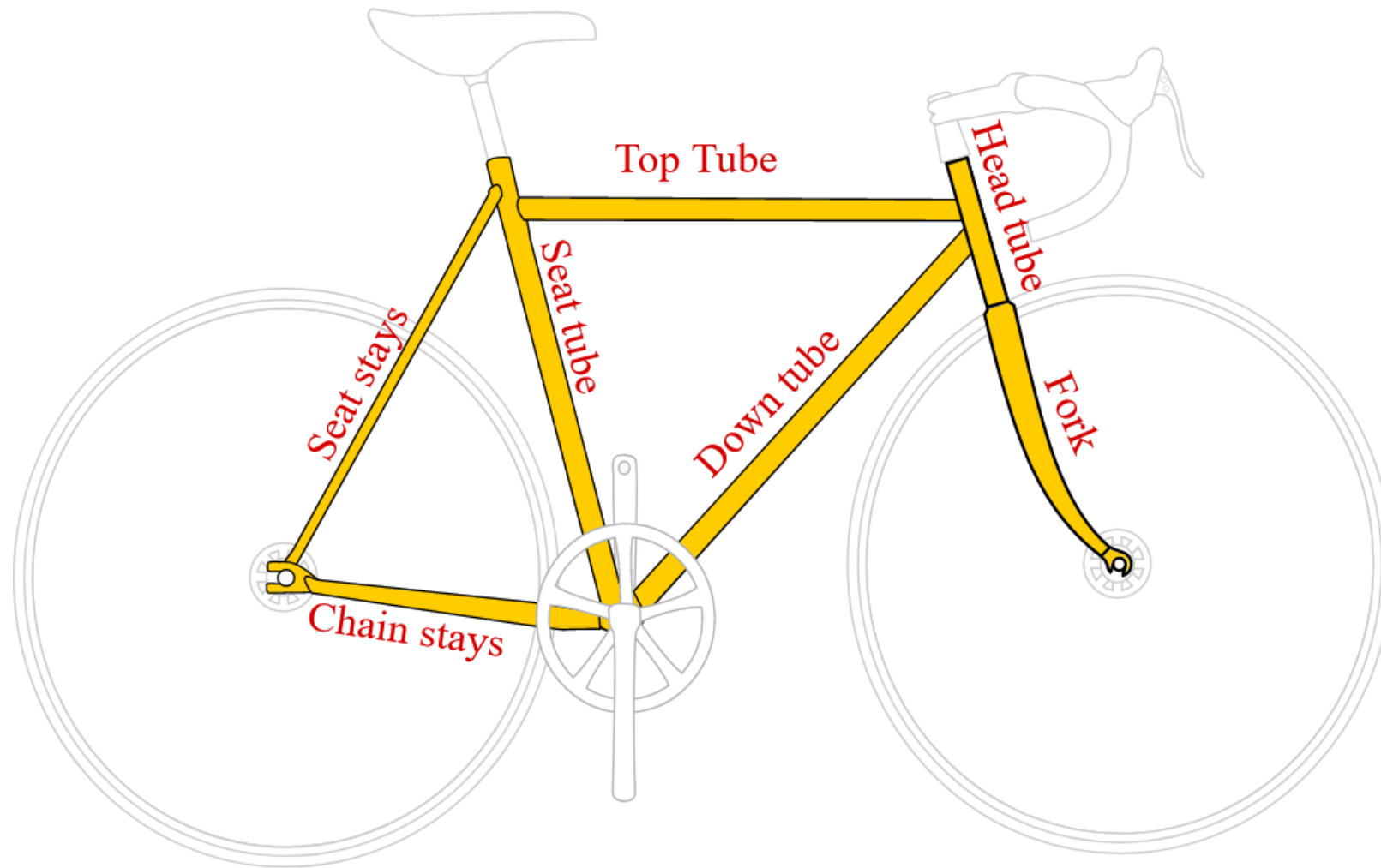
MATERIALS SELECTION CRITERIA

- Performance requirements
- Durability and reliability requirements
- Mass and volume requirements
- Cost of production
- Compatibility with manufacturing processes
- Industrial standards
- Government regulations
- Intellectual property requirements
- Biocompatibility
- Aesthetic properties

BEHAVIOUR, PROPERTIES AND PERFORMANCE CHARACTERISTICS OF ENGINEERING MATERIALS



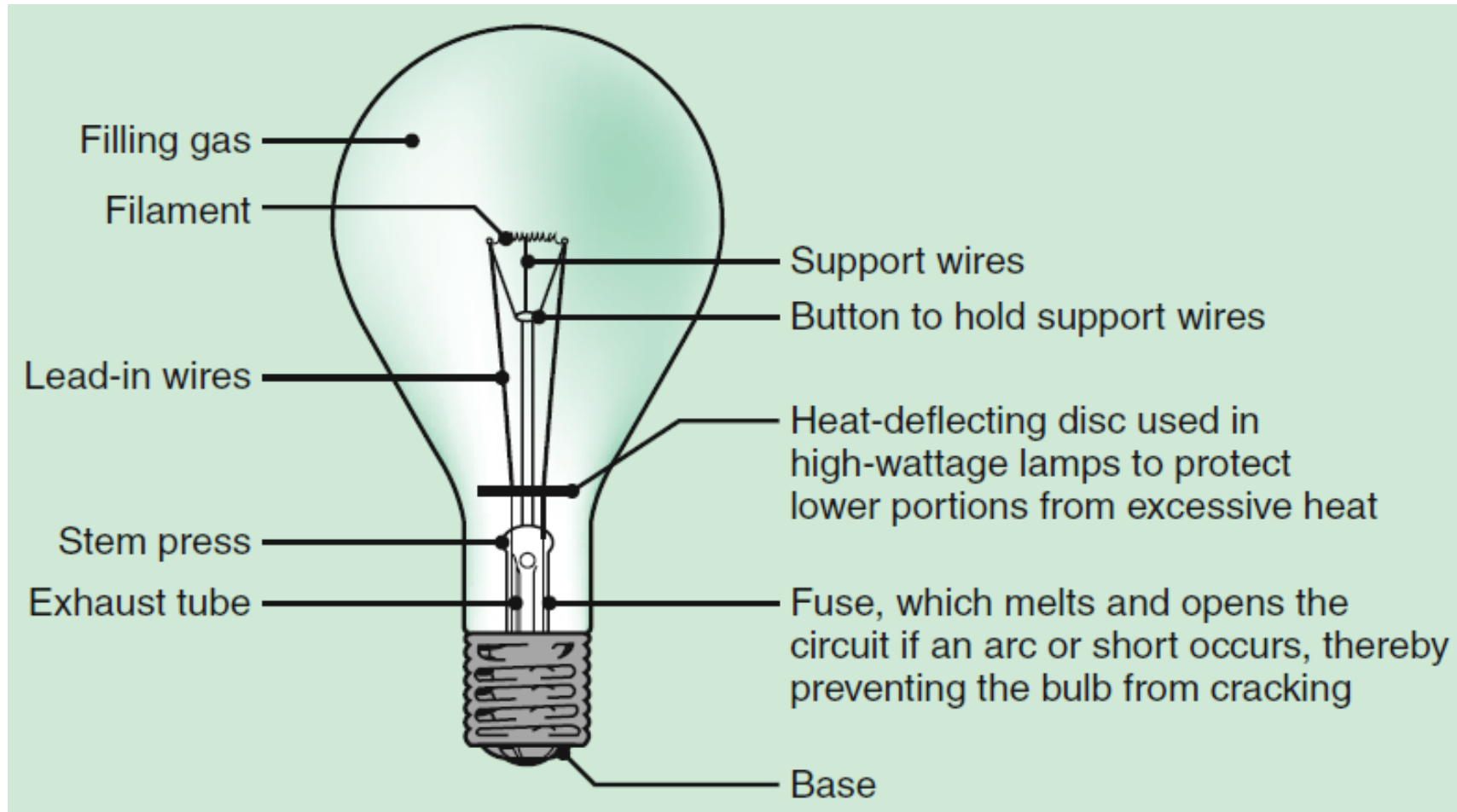
WHAT IS THE BEST MATERIAL FOR A BICYCLE FRAME?



Components of a bicycle frame

- Weight
- Strength
- Stiffness
- Manufacturability
- Cost
- Corrosion
- Aesthetics

WHAT IS THE BEST MATERIAL FOR AN INCANDESCENT LIGHT BULB?

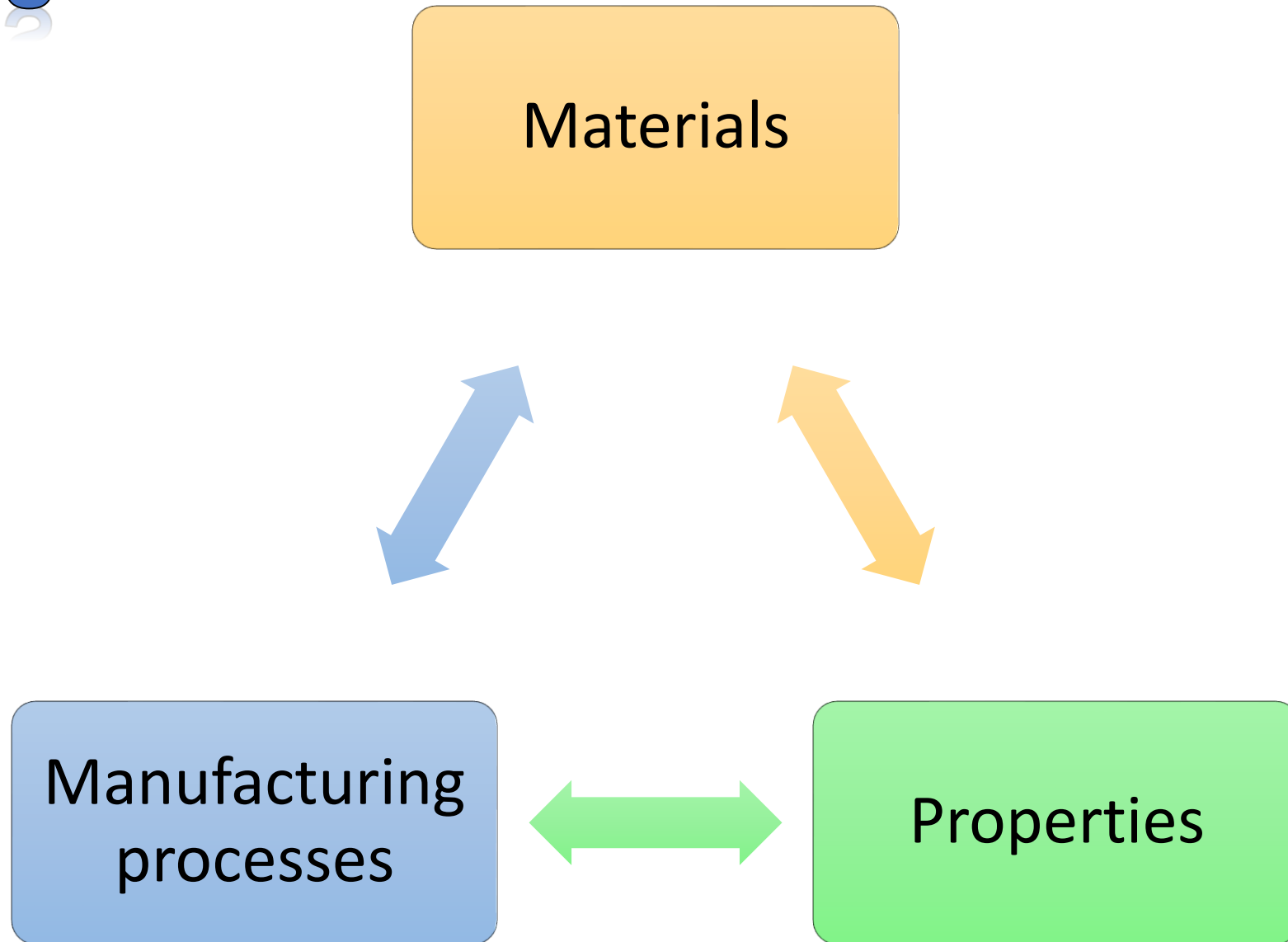


Components of an incandescent light bulb

- Electrical conductivity
- Thermal conductivity
- Thermal expansion
- Melting point
- Oxidation
- Strength

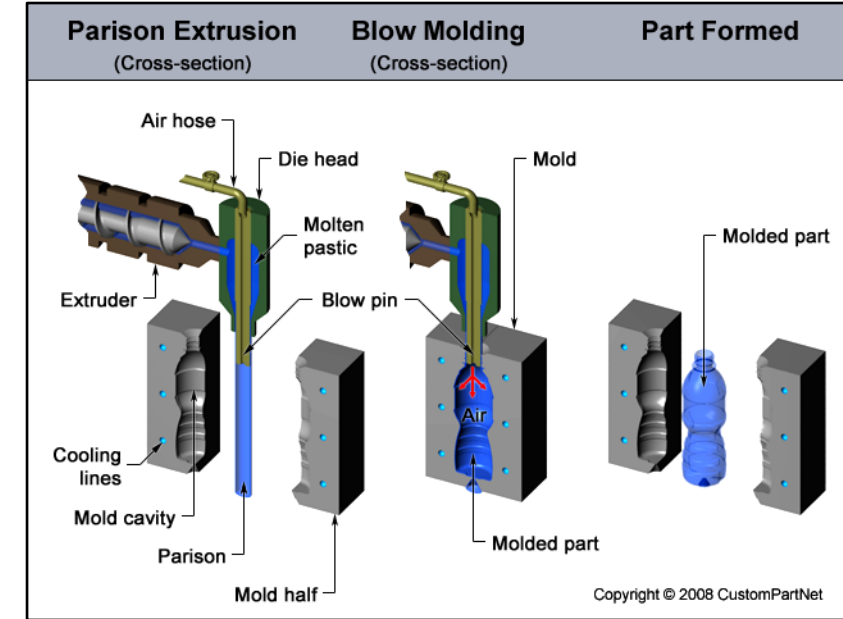
RELATIONSHIP BETWEEN MATERIALS, PROPERTIES AND MANUFACTURING PROCESSES

MATERIALS, PROPERTIES AND MANUFACTURING METHODS

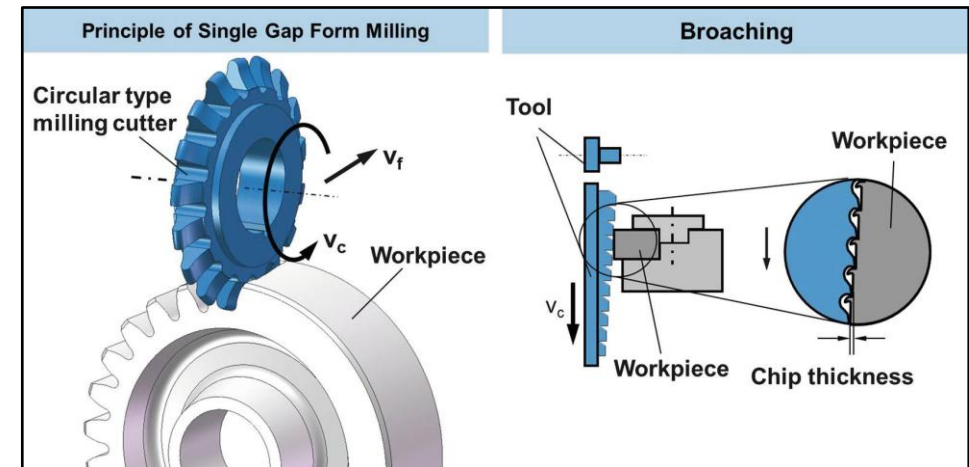


MANUFACTURING PROCESS ↔ MATERIALS

- How plastic boxes are made?
 - Plastics can be easily melted and injected into a mould cavity
 - Bottles, lunch boxes, laptop housings, and almost all the plastic casings are made using molten plastics → **Injection moulding**
- How gear wheels are made?
 - A metal piece is cut into the shape of the gear → **Machining**
 - Material removal process
 - Nuts, bolts, screws, pistons



Blow moulding process



Gear machining process

MANUFACTURING PROCESS ↔ PROPERTIES

- How a knife is made?
 - Metal part is usually **forged** (heated and hammered into a shape of a knife) → Improve strength
 - Cutting edge of the knife is **quenched** (heated and rapidly cooled by immersing in water or oil) → Improve hardness



Knife blade

MATERIAL FAILURE

DIFFERENT WAYS OF MATERIAL FAILURE

- Ductile fracture
- Brittle fracture
- Fatigue failure
- Creep failure
- Corrosion
- Wear
- Failure due to residual stresses
- Thermal cracking



Ductile fracture



Corrosion of a pipe flange

Kalpajian, S., & Schmid, S. R. (2006). Manufacturing Engineering and Technology (6th ed.). Pearson Education South Asia.
<https://gfycat.com/bonyimpressiveinchworm>
<https://www.alliedcorrosion.com/cathodic-protection-basics/basic-theory-of-metallic-corrosion/>

COST OF MATERIALS

COST OF MATERIALS

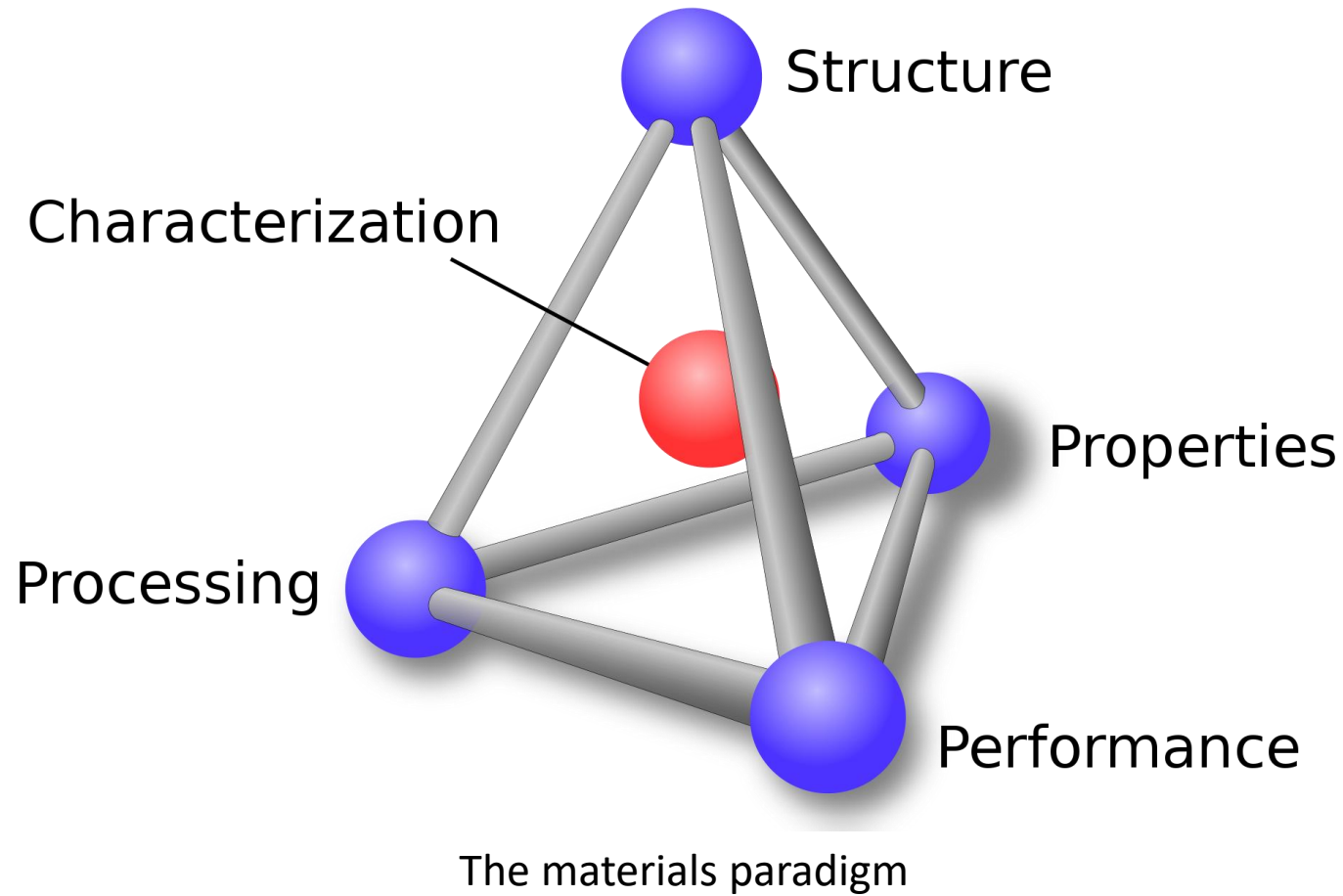
- Typical cost breakdown in manufacturing
 - Design
 - 2D drawings, 3D models, computational analyses
 - Materials (**~50%**)
 - Raw materials, initial material processing, abundance
 - Tooling
 - Cutting tools, dies, moulds, work-holding devices, fixtures
 - Fixed
 - Energy, rentals, taxes
 - Capital
 - Machines, equipment, lands, buildings
 - Labour
 - Machinists, helpers, servicing personnel



Different types of cutting tools

We are at the end of this section

- Now you should be able to
 - LO-6: Use of basic hand tools and machine tools in engineering workshop for machining processes and joining metals
 - **Materials used for such manufacturing processes**



End